Abstract

Artificial Intelligence (AI) has become deeply integrated into modern economies, but its reliance on energy-intensive data centers raises environmental concerns. As AI continues to be adopted, its associated electricity and water demand, greenhouse gas (GHG) emissions, and impacts on the urban heat island (UHI) effect may become more pronounced, with potential implications for public health and urban sustainability. This raises questions about the capacity of existing policy frameworks. This paper examines San Francisco as a case study due to its position as a hub of AI development. Specifically, it analyzes San Francisco's recently adopted Climate Action Plan and the implications of recent California legal efforts, including the vetoed SB 1047 and recently passed SB 53, while highlighting the pending SB 886 and SB 1168. The study evaluates the extent to which these legislative frameworks address the emerging environmental consequences across policy areas such as transparency, accountability, and oversight. Ultimately, this paper explores the tradeoffs policymakers face between supporting technological innovation and long-term sustainability goals, arguing that current governance frameworks may not yet fully account for the environmental realities of AI expansion.

Introduction

Artificial Intelligence (AI) has rapidly transformed industries by automating tedious corporate tasks through scheduling, data analysis, and information processing. AI systems are powered by data centers, which require substantial amounts of electricity to operate and cool. Recent projections from the International Energy Agency estimate global electricity demand from data centers to continue rising in coming years as AI becomes more widely adopted ("Energy Demand," 2025).

As concerns over the consequences of climate change intensify, the environmental impacts of AI infrastructure—including public health, emissions, and grid strain—have raised questions about the role of policy frameworks in regulating AI-driven growth. Researchers have warned about data centers' contributions to heat generation in surrounding communities (Marinoni et al., 2026).

This paper is a case study on San Francisco, a major center of AI development and investment. Currently, AI continues to embed itself across San Francisco's economy, powering everything from customer service to logistics (Redmond, 2026). With cities like San Francisco, debates are emerging over governmental regulation of AI and its associated impacts.

Recent legislation in California reflects ongoing debate over the appropriate stringency of AI disclosure requirements, including the Safe and Secure Innovation for Frontier Artificial Intelligence Models Act (SB 1047) and later the Transparency in Frontier Artificial Intelligence Act (SB 53). SB 1047 was vetoed, reflecting concerns over aggressive regulation's potential impacts on innovation and growth (Allyn, 2024). Policymakers have also tried to limit data centers' energy usage through the pending California Technology Innovation and Ratepayer Protection Act (SB 886) as well as the Data Centers: Rate Structures bill (SB 1168).

The purpose of this case study is to evaluate current policy gaps over AI-related energy consumption and examine a variety of approaches that governments could use to balance environmental sustainability with technological innovation.

Energy Costs of AI

Data centers power AI systems and are responsible for a substantial and growing share of electricity demand. Electricity consumption from data centers in 2025 was around 415 terawatt hours (TWh), representing an annual growth of 12% over the previous five years ("Energy

Demand," 2025). To accommodate the energy demands of AI-driven growth, cities like San Francisco have become more power-dense.

This trend is projected to continue in coming years. Figure 1 shows a Base Case analysis, where consumption is expected to double to around 945 TWh by 2030 and grow annually by around 15% between 2030 and 2040. This is more than four times faster than the combined growth in other sectors ("Energy Demand," 2025). This growth rate raises the question of whether disclosure-based policy can keep pace.

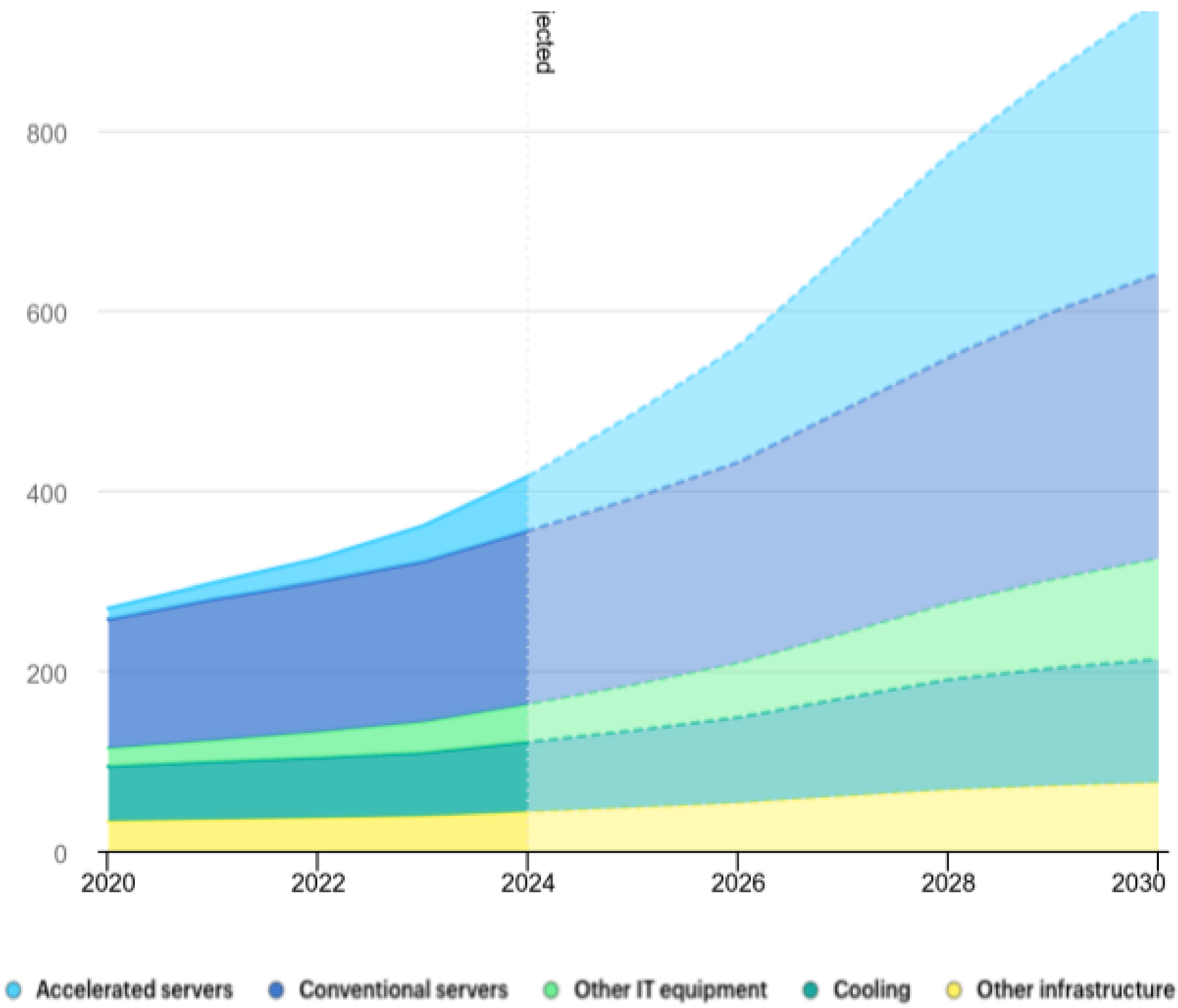

Figure 1
Global Data Centre Electricity Consumption, by Equipment, Base Case, 2020–2030. From *Energy and AI*, by International Energy Agency, 2025, IEA, Paris (iea.org). CC BY 4.0.

One reason for AI's high energy demand lies in its continuously-operating data centers. Around 60% of the energy used by data centers is to power their servers and process digital information. The most advanced servers perform trillions of calculations every second, requiring two to four times as many watts to run compared to their traditional counterparts (Leppert, 2025). As the technology advances, growing concerns about these energy-intensive systems are increasingly prompting calls for regulation.

Another reason for energy usage is the data centers' cooling systems. Since they carry out so many operations in a short timespan, they require energy-intensive cooling systems to prevent overheating. This process requires a significant amount of water, where in 2023 the country's data centers consumed around 17 billion gallons of water (Leppert, 2025). Furthermore, total water consumption is projected to increase 4.6-fold, reaching anywhere from 74.54 to 116.63 billion liters between the years 2019 and 2028 (Liu et al., 2025). This poses challenges for data centers in California, where water conservation is a necessary consideration due to the state's dry conditions.

Public Impacts of AI Infrastructure

Data centers' massive carbon footprints have negatively affected local communities' public health. They rely on backup generators, which emit particulate matter, sulfur dioxide, nitrogen oxides, and volatile organic compounds into surrounding neighborhoods. Exposure to these pollutants is commonly associated with cardiovascular disease (Liu et al., 2025). As temperatures continue to rise, electricity dependence may increase and consequently heighten energy strain. The resulting feedback loop would simultaneously increase energy requirements, greenhouse gas (GHG) emissions, and associated health risks.

The rise in temperatures can be further explained by data centers' contributions to the urban heat island (UHI) effect. The UHI effect occurs when cities replace natural vegetation with heat-absorbing infrastructure, resulting in elevated temperatures compared to surrounding regions. Figure 2 demonstrates that over the span of a year, AI data centers in urban cities have contributed to a temperature increase between 1.5 °C and 2.4 °C (Marinoni et al., 2026).

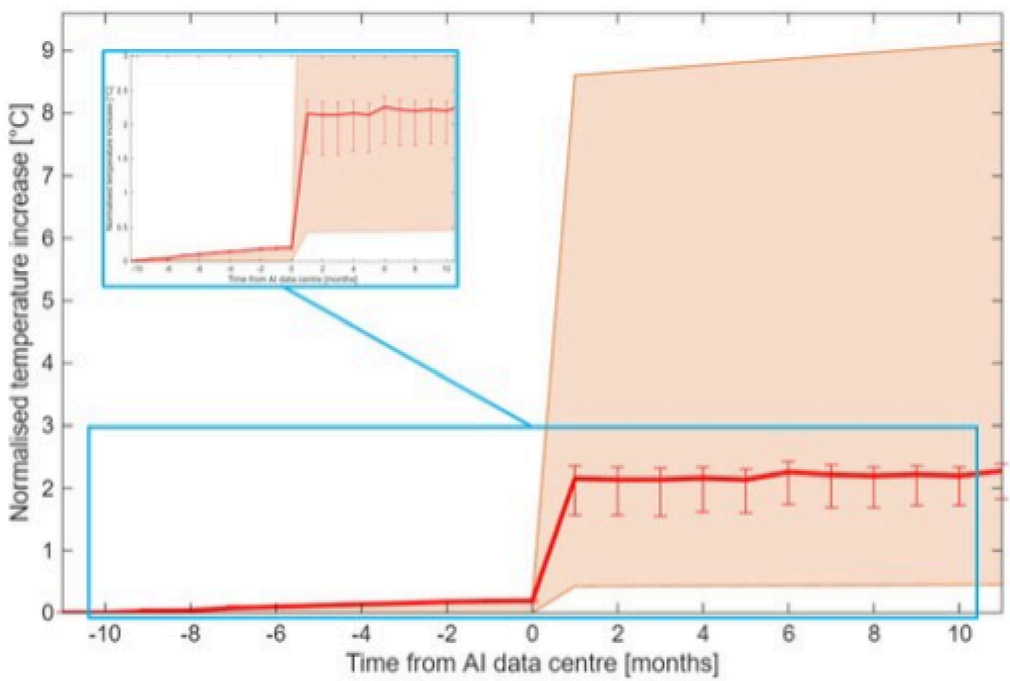

Figure 2

Temperature Increase Through Time Over the AI Hyperscalers Locations Centred Around the Time of Start of Operations. Reproduced from Marinoni et al. (2026) under a CC BY-NC-ND 4.0 license.

One factor of the UHI effect is data centers' proximity to one another. Figure 3 demonstrates that the average temperature increase can be measured up to 4.5 kilometers away from a data center, although the effect seems to reduce to 30% intensity when the distance expands to 7 kilometers (Marinoni et al., 2026). This is particularly relevant to cities neighboring San Francisco, where the concentration of data centers has been attributed to documented rises in local temperature (Wolverton, 2025).

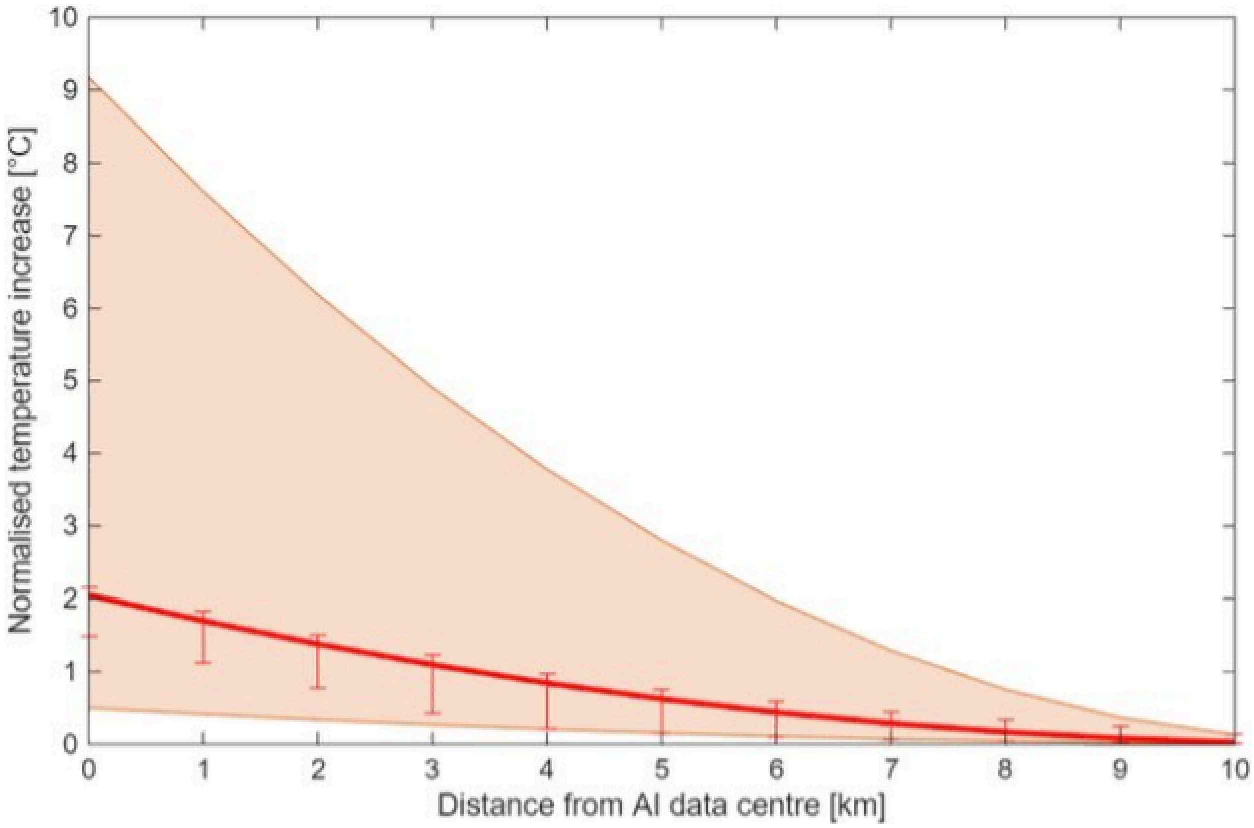

Figure 3

Temperature Increase Through Space as a Function of the Distance from the AI Hyperscalers Locations. Reproduced from Marinoni et al. (2026) under a CC BY-NC-ND 4.0 license.

Policy Gap

San Francisco has publicly declared through its 2026 Climate Action Plan that cutting down its GHG emissions would remain a top priority. The plan includes a comprehensive explanation for how the city plans to achieve net-zero by 2040 through first identifying the main contributors to existing emissions.

The city classifies its carbon output through three distinct emission scopes, as outlined by the GHG Protocol. Scope 1 classifies direct carbon output from sources owned and controlled by a corporation. Scope 2 and Scope 3 are indirect, since they refer to emissions generated from a purchased source both upstream and downstream the value chain (World Resources Institute & World Business Council for Sustainable Development, 2004). AI falls under Scope 3, since the energy required to power its data centers is often sourced in regions external to the corporation and city.

Even though the Climate Action Plan is thorough in its analysis, it leaves out AI energy consumption as a sustainability concern (Redmond, 2026). The prevalence of AI-related emissions thus runs contrary to the city's goals of carbon neutrality. Furthermore, the Climate Action Plan's failure to engage with the debates surrounding SB 1047 and SB 53 highlights growing concerns about regulatory policies' potential tradeoffs between innovation and sustainability.

SB 53 was signed into law in September 2025. The regulatory bill follows SB 1047 and aims to impose a secondary level of protection for whistleblowers. Both bills mandate corporate transparency, but SB 53 is a frontier-AI safety transparency bill requiring detailed security insights and reports before a product is released.

The policy gap lies in governments' inability to calculate the exact environmental footprint that AI leaves given that its energy consumption is mediated by complex supply chains and electricity generation. This makes SB 53's reporting difficult, since it offers no way to consolidate Scope 3 information from suppliers and partners, resulting in inconsistent data structures and measurement methodologies. Additionally, such a bill would place developers in legal jeopardy, as they are required to disclose internal decisions in significant detail. Furthermore, SB 53's reliance on a 10^{26} FLOPs computational threshold for "large" models is arbitrary and incapable of capturing individual nuances (Hizkias, 2025), a critique advanced by the Chamber of Progress, a tech-industry-funded trade group advocating against stringent AI regulation.

Besides disclosure, there are two pending bills that pose financial and design barriers for corporations utilizing data centers: SB 886 and SB 1168. SB 886 mandates that data centers pre-fund fifteen-year contracts for new zero-carbon energy infrastructure so that utility ratepayers do not shoulder the financial burden of AI expansion. SB 1168 would levy heavy fines on data centers for excessive energy consumption, offsetting residential electricity costs.

Although both bills are aggressive attempts to regulate AI energy, they suffer from exploitable limitations. SB 886's high applicability threshold of 25 megawatts allows companies to avoid penalties by breaking up energy loads into clusters of smaller, decentralized data centers in close proximity, amplifying the UHI effect. As for SB 1168, the fines may be seen as an insufficient deterrent that is a mere "cost of doing business" rather than a true consumption ceiling, allowing large technology companies to absorb fines without meaningfully changing behavior. Ultimately, the fact that both bills apply only to infrastructure within California means that they cannot prevent corporations from moving AI operations to neighboring states with weaker oversight.

Policy Options and Tradeoffs

To address the disclosure gap, more standardized reporting requirements for AI developers and data center operators can be created. Instead of having various methodologies for disclosure, there could be common metrics for electricity consumption, water usage, and GHG emissions. Accurate reports would help cities like San Francisco better frame AI's environmental impacts within existing climate planning efforts.

However, detailed reporting might impose compliance costs on smaller firms and still carry over previous concerns about disclosing private operational information. Therefore, policymakers could consider how to balance accountability with avoiding disproportionate compliance burdens on emerging companies.

Even with more specific disclosure policies and grid-focused consumption taxes like SB 1168, policymakers should direct attention to data centers' environmental and public health impacts. Since the UHI effect and health consequences are associated with their waste exhaust, additional regulatory attention should be directed towards their power systems. Traditional backup systems are powered by diesel generators, which worsen air quality by emitting pollutants (Liu et al., 2025). Finding alternative ways to supply power—such as massive battery storage or green hydrogen fuel cells—can reduce conventional diesel systems' harms. This is particularly critical as grid-vulnerability bills like SB 886 penalize high grid-draws, unintentionally encouraging developers to rely more heavily on localized backup infrastructure.

Transitioning towards cleaner backup power can be difficult, since alternative energy requires significant upfront investment. This could consequently increase the burden on smaller operators during adoption. Backup system replacements must also deliver consistent power under unforeseen circumstances while meeting baseline data center demand. Policymakers could balance environmental objectives with concerns over cost, energy reliability, and overall feasibility. After all, the extent of these technologies' benefits ultimately hinges on reliable resource availability.

An additional policy action could be the incorporation of AI-infrastructure growth into future climate planning. As AI continues to expand, governments could benefit from acknowledging its influence on sustainability efforts within the Climate Action Plan, allowing policymakers to complement technological innovation with responsive environmental solutions.

However, planning can be difficult due to the technology's volatile and unpredictable growth, potentially affecting the accuracy of sustainable growth projections. AI predictions routinely fail not because the technology lacks potential, but because forecasters tend to model AI as an isolated technical artifact rather than a socio-technical system shaped by institutions, regulation, and human behavior (Thevaranjan, 2026). This error causes projections to overestimate short-term impacts while misreading longer-term ones, potentially complicating any policy effort premised on accurate growth assumptions. Even though these projections may be imperfect, aligning policy with status quo realities is a necessary next step.

Conclusion

Artificial intelligence has become an increasingly important driver of economic growth and innovation, but its expansion has also introduced new environmental and health challenges through greenhouse gas emissions and the urban heat island effect.

These concerns are applicable to San Francisco, a city that is trying to progress toward a net-zero future yet experiences elevated demand for AI. This case study suggests that San Francisco's current frameworks like the Climate Action Plan and California's recent legislation represent important steps toward accountability yet still contain structural limitations. Difficulties in measuring indirect Scope 3 emissions combined with concerns over aggressive regulation suggest that there is more nuance than a one-size-fits-all solution. This complexity extends to potential approaches, including unified transparency, cleaner energy infrastructure, and accurate climate planning.

With AI's rapid adoption, the question becomes whether environmental policy can adapt quickly enough to sufficiently manage potential consequences. Ensuring that climate planning and governance frameworks accurately represent a changing status quo will be essential for long-term sustainability.

References

Allyn, B. (2024, September 29). *California Gov. Newsom vetoes AI safety bill that divided Silicon Valley*. NPR. https://www.npr.org/2024/09/20/nx-s1-5119792/newsom-ai-bill-california-sb1047-tech

Energy demand from AI. (2025). International Energy Agency. https://www.iea.org/reports/energy-and-ai/energy-demand-from-ai

Hizkias, A. (2025, July 9). *Why California's SB 53 still gets AI regulation wrong*. Chamber of Progress. https://progresschamber.org/insights/why-californias-sb-53-still-gets-ai-regulation-wrong/

Leppert, R. (2025, October 24). *What we know about energy use at U.S. data centers amid the AI boom*. Pew Research Center. https://www.pewresearch.org/short-reads/2025/10/24/what-we-know-about-energy-use-at-us-data-centers-amid-the-ai-boom/

Liu, R., Wu, Z., & Ren, S. (2025, November). *An assessment of California data centers' environmental and public health impacts* (N. Perry, M. E. Perry, S. Leonard, & G. Ochs, Ed.). Next 10. https://www.next10.org/sites/default/files/2025-11/ai-environmental-public-health-costs.pdf

Marinoni, A., Cambria, E., Lin, W., Mura, M. D., Chanussot, J., Ragusa, E., Tso, C. Y., Zhu, Y., & Horton, B. (2026). The data heat island effect: Quantifying the impact of AI data centers in a warming world. *ResearchGate*. https://doi.org/10.48550/ARXIV.2603.20897

Redmond, M. (2026, March 27). San Francisco's climate change goals don't account for the AI boom. *San Francisco Chronicle*.

Thevaranjan, H. (2026, February 11). *We keep getting AI wrong and history keeps telling us why* Medium. https://medium.com/data-science-collective/we-keep-getting-ai-wrong-and-history-keeps-telling-us-why-97180ee7d351

Wolverton, T. (2025, June 27). *Location of state's data centers linked to polluted areas*. San Francisco Examiner. https://www.sfexaminer.com/news/technology/california-data-centers-concentrated-in-polluted-areas/article_a4721c55-1782-4d27-9e43-84cad7a2fb5f.html

World Resources Institute & World Business Council for Sustainable Development. (2004). *The greenhouse gas protocol: A corporate accounting and reporting standard (Revised ed.)*. Greenhouse Gas Protocol. Greenhouse Gas Protocol Report